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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
VI Semester B. E. Regular Examinations Sept/Oct– 2024
Information Science Engineering
INFORMATION RETRIEVAL (ELECTIVE)

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

- Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
- Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

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1	1.1	Draw the block diagram depicting view of a document: from full text to asset of index terms.	02	1	1
	1.2	Let U be the universe of discourse, A and B be two fuzzy subset of U . $\mu_{A \cup B}(u)$ is given by $\max(\mu_A(u), \mu_B(u))$.	01	1	1
	1.3	List any two practical issues of Information Retrieval.	01	2	1
	1.4	$q_{dnf} = [(1,1,1) \vee (1,1,0) \vee (1,0,0)]$ is the disjunctive normal form of <u>Boolean</u> . (Each component is binary weighted vector associated with tuple (k_a, k_b, k_c)).	01	3	1
	1.5	An IR model is a quadruple $[D, Q, F, R(q_o, d_j)]$. Here F indicates <u>framework for modeling of I</u> .	01	1	1
	1.6	Diagrammatically represent the characteristics of a document.	01	2	1
	1.7	A Boolean query which selects all documents which satisfy e_1 <u>OR</u> e_2 eliminating duplicates.	01	2	2
	1.8	<u>heaps</u> law implies that the length of the words in the vocabulary increases logarithmically with the text size.	01	1	1
	1.9	<u>User relevance</u> feedback is the most popular query reformation strategy.	01	1	1
	1.10	A high <u>novelty</u> ratio indicates that the system is revealing many new relevant documents which were previously unknown.	01	1	1
	1.11	Identify any two Browsing models.	01	1	1
	1.12	The cost of algorithm of merging the partial indices in a binary fashion is given by the big O notation _____.	01	3	1
	1.13	What is the purpose of two sets of tags in Mathematical Markup Language? <u>presentation and content</u>	01	3	1
	1.14	The backward DAWG matching algorithm is based on _____ automaton.	01	1	1
	1.15	_____ has been used to parallelize the computation of the dynamic programming matrix.	01	1	2
	1.16	Write the related established by Amdahl's law in the form of inequality relation.	01	1	4
	1.17	The most accurate technique for merging ranked hit-lists is to use _____.	01	1	1
	1.18	Draw the typical Crawler-indexer architecture.	02	1	2

PART-B

2	a	Compare Information Retrieval and Data Retrieval.	04	3	1
	b	Describe the process of retrieving information.	04	2	2

c		Explain the following: i) Vector model ii) Probabilistic mode for IR system.	08	2	1			
3	a	Differentiate between Belief network model and Inference network model.	04	2	1			
	b	Assume the following: A database contains 80 records on a particular topic. A search was conducted on that topic and 60 records were retrieved. Of the 60 records retrieved, 45 were relevant. Compute the precision and recall scores for the search.	06	3	4			
	c	Describe the following with respect to retrieval evaluation performance: i) E-measure ii) User-oriented measure.	06	1	1			
		OR						
4	a	List and explain the different types of pattern matching.	06	2	2			
	b	List and draw the three main structures in structural queries.	03	2	1			
	c	Explain any four query protocols.	04	2	1			
	d	Write the six steps of “complete link algorithm”.	03	1	2			
5	a	Outline the five text operations adopted in document preprocess procedure.	05	2	1			
	b	Explain the three general steps followed in search algorithm on an inverted index.	03	2	1			
	c	Illustrate Shift-OR algorithm with respect to indexing and searching in IR system.	05	2	2			
	d	Plot the graph showing “Tradeoff of index space versus word searching time”.	03	3	3			
		OR						
6	a	Explain with an example the backward <i>DAWG</i> matching algorithm.	05	2	3			
	b	Draw the table depicting the comparison of text compression techniques.	03	1	1			
	c	Explain the three parts of HyTime architecture. Also mention one of its applications.	05	2	3			
	d	Discuss the concept of entropy with relevance to Information Theory.	03	2	4			
7	a	Develop the parallel inverted file consisting tables of postings, index and posting tables for the below mentioned documents. Also provide the tables for the following: i) Skewed and partitioned postings ii) Partitioned postings files. Documents <table><tr><td>This little piggy went to market.</td><td>This little piggy stayed home.</td><td>This little piggy had roast beef.</td></tr></table>	This little piggy went to market.	This little piggy stayed home.	This little piggy had roast beef.	08	4	4
This little piggy went to market.	This little piggy stayed home.	This little piggy had roast beef.						
	b	List and explain the key problems posed by web related to data.	05	2	3			
	c	Briefly explain web query languages.	03	2	3			
		OR						
8	a	Discuss the concepts of the following: i) Gatherers and brokers in Harvest architecture. ii) Crawler-indexer architecture.	08	2	3			

	b	Summarize the concepts associated with distributed <i>IR</i> .	03	2	4
	c	List the four classes in the taxonomy of parallel computing. With a block diagram, explain partitioned processing on <i>MIMD</i> machine.	05	1	3
9	a	Describe the InfoGrid system with relevant block diagrams.	08	2	3
	b	Explain automatically derived collection overviews with respect to user interface and visualization.	05	2	3
	c	List the four cases in studies of user interactions with relevance feedback system.	03	1	4
OR					
10	a	Draw the flowchart representing simplified model of the information access process.	03	2	1
	b	Explain the role of visualization from information retrieval perspective.	05	2	3
	c	Explain the following with neat diagrams: i) InfoCrystal retrieval result display ii) <i>VIBE</i> retrieval result display.	08	2	3